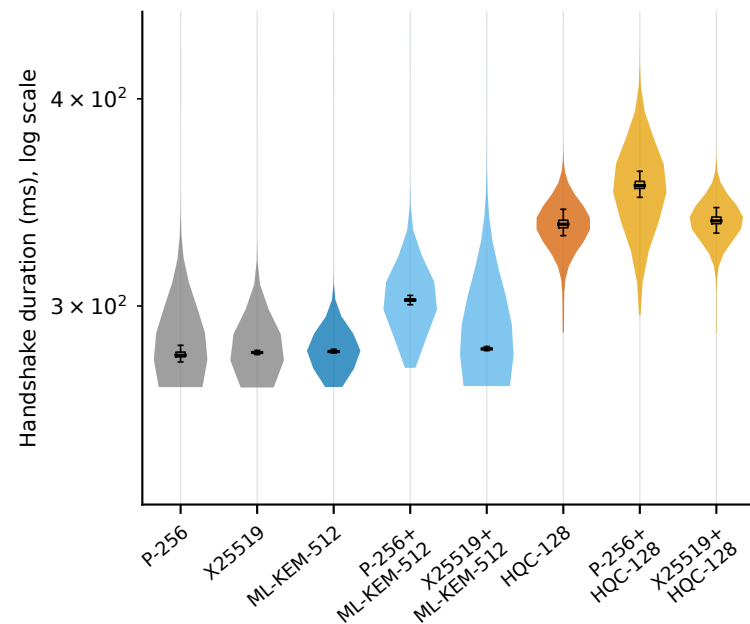
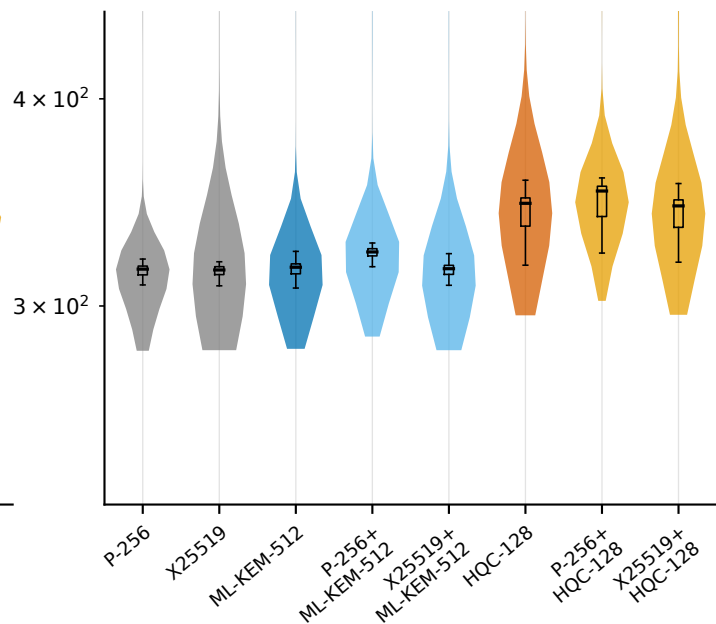


(a)

ed25519 (classical)

**(b)**

mldsa44 (post-quantum)

**(c)**

Classical KEX ML-KEM (pure) ML-KEM (hybrid) HQC (pure) HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	mldsa44 Mean (ms)	mldsa44 CV (%)	mldsa44 Max (ms)
P-256	285.5	23.3	1.30k	314.0	14.2	1.12k
X25519	283.5	16.0	1.29k	319.8	31.7	1.62k
ML-KEM-512	283.2	10.8	952.1	315.5	18.8	1.58k
P-256+ML-KEM-512	304.7	16.6	1.42k	323.1	18.5	1.62k
X25519+ML-KEM-512	289.8	31.1	1.45k	317.0	23.5	1.59k
HQC-128	338.0	10.4	840.4	349.4	28.6	1.79k
P-256+HQC-128	360.4	19.5	1.67k	349.5	18.7	1.77k
X25519+HQC-128	340.3	9.2	843.4	347.6	26.6	1.77k